PREP 3, LESSONS 5–6

Radioactive decay and nuclear equations

Unstable nuclei; activity and count-rate; alpha, beta and gamma; balancing equations

SCORE / 10

Name _____ Class _____ Due _____

Do Now REMEMBER

From earlier lessons, 1 mark each

1 Name the model of the atom with electrons embedded in a positively charged sphere. DN 246 [1]

The plum pudding model

2 State how an electron can move to a higher energy level. DN 235 [1]

By absorbing electromagnetic radiation.

3 State the type of ion formed when an atom loses one or more electrons. DN 717 [1]

A positive ion

Key ideas LESSONS 5–6

Show your working

4 The activity of a source is measured in **becquerels (Bq)** / counts / joules (J) [1]

5 A detector is placed near a radioactive source. Explain why it records only some of the radiation the source emits. [1]

Radiation leaves the source in all directions, so most of it misses the detector ✓

6 An alpha particle is **two protons and two neutrons** / a high-speed electron / electromagnetic radiation [1]

7 Complete the nuclear equations. Mass numbers and atomic numbers must balance.

a Alpha decay: $^{238}_{92}U \rightarrow \underset{90}{\overset{234}{\square}}Th + {}^{4}_{2}He$ 234 ✓ 90 ✓ [2]

b Beta decay: $^{14}_{6}C \rightarrow \underset{7}{\overset{14}{\square}}N + {}^{0}_{-1}e$ 14 ✓ 7 ✓ [2]